

Delhi Public School, Bokaro Steel City

Class 10 Biology · Unit Test

1. Which blood vessel carries blood away from the heart? [2]
2. Which of the following organelles is primarily involved in photosynthesis? [2]
3. Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outlining the two major stages of the process. [5]
4. Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta, include the names of the valves crossed. [5]
5. Draw a labelled diagram of an alveolus showing capillaries and air space (label all exchange). [3]
6. Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine, liver, pancreas) and label the site where most absorption occurs. [5]
7. Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule, collecting duct). [4]

- 8.** Explain the structural differences between palisade mesophyll and spongy mesophyll and state how each structure aids its function in the leaf. [3]
- 9.** Describe the process of transpiration in plants in two to three sentences and name two environmental factors that increase its rate. [2]
- 10.** Explain how the structure of xylem vessels facilitates water transport in plants (mention one adaptation and explain its role). [2]
- 11(a).** A diagram shows two potted plants — Plant A in bright light with high water needs, Plant B kept in dim light with pale, elongated leaves. [2]
- 11(b).** Suggest one practical measure to help Plant B recover. [1]
- 12.** A resting person has tidal volume (air per breath) of 0.5L and breathes 12 times per minute. [2]
- 13.** If dead space is 0.15L per breath, calculate alveolar ventilation per minute. Show working. [2]